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Effect of automobile pollution on Chlorophyll content of road side herbs along NH-31 from Katihar to Purnea (Bihar)

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Abstract- Automobile exhaust contains a large no. of air pollutants such as carbon monoxide, unburnt hydrocarbons, Sulphur and nitrogen oxide, tetraethyl lead and tetramethyl lead and several other lead compounds. These air pollutants adversely affect road side herbs by decreasing their chlorophyll content. In the present study, chlorophyll content of 10 herbs (*Catharanthus roseus*, *Ageratum conyzoides*, *Tridax procumbens*, *Acalypha indica*, *Phyllanthus nururi*, *Parthenium hysterophorus*, *Croton sparsiflorus*, *Euphorbia hirta*, *Emilia sonchifolia*, *Phyla noduliflora*) was estimated in road side plants and countryside plants i.e. one km away from the road side in summer season. The chlorophyll contents of road side herbs were less than those of countryside herbs.

Key words: Automobile exhausts, carbon monoxide, unburnt hydrocarbon, sulphur & nitrogen oxides, lead compounds

INTRODUCTION

Air pollution in India is quite a serious issue with the major sources being fuel wood and biomass burning, fuel adulteration, vehicle emission and traffic congestion. In autumn and winter months, large scale crop residue burning in agriculture fields - a low cost alternative to mechanical tilling - is a major source of smoke, smog and particulate pollution. India has low per capita emissions of greenhouse gases but the country as a whole is the third largest after China and the United States. Transport is the dominant source of emissions of Sulphur dioxide (SO_2), 57% of oxides of Nitrogen (NOx), 80% of lead (pb) 81% of Carbon monoxide (CO) and 75% of hydrocarbon (HC) emissions.[1]

A number of countries have targeted vehicles and associated sectors (such as fuel) to overcome the problem of air pollution. Notable successful initiatives are: conversion of public transport from diesel to CNG, switching Vikrams (tuk-tuks) from diesel to electricity, shifting from leaded to unleaded gasoline in many countries etc. Still the pollution problem in urban cities is continued. The Air (Prevention and Control of Pollution) Act was passed in 1981 to regulate air pollution. Automobile exhaust emissions contribute significant air pollution.[2-4]

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MATERIAL & METHOD

The study area was selected NH-31 from Katihar to Purnea, a total distance of 30km. 10 common herbs- *Catharanthus roseus*, *Ageratum conyzoides*, *Tridax procumbens*, *Acalypha indica*, *Phyllanthus nururi*, *Parthenium hysterophorus*, *Croton sparsiflorus*, *Euphorbia hirta*, *Emilia sonchifolia*, *Phyla noduliflora* were selected for study. All selected herbs were collected in summer season. Herbs one km. away from road side were also collected.

Estimation of Chlorophyll:

Chlorophyll content of leaves from each plant was estimated in laboratory. 250 mg of fresh leaves were cut into small pieces and transferred in 25 ml 80% acetone. It was incubated in dark for 24 hrs. absorbance at the extract was taken at 645 nm and 663 nm with spectrophotometer. Chlorophyll was estimated by the formula.[5]

$$\text{Total Chlorophyll}\left(\frac{\text{mg}}{\text{gm}}\text{Fw}\right) = \frac{20.2(A645) + 8.02(A663)}{1000X\ Fw} X\ V$$

A_{645} = Absorbance at 645 nm
A_{663} = Absorbance at 663 nm
V = Volume of Chlorophyll extract
Fw = Weight of fresh leaves

RESULT

In the present study, 10 herbs belonging to family Euphorbiaceae, 4 from family Compositae, 1 from Apocyanaceae and 1 from Verbinaceae were collected from road side of NH-31 (Katihar to Purnea) in summer season. Same herbs were also collected from one km away of road side. Chlorophyll pigment was estimated from each herb.

It was observed that the chlorophyll content in each herb of road side decreased in comparison to herbs from one km away of road side. The result is tabulated in table 1.

The chlorophyll content was highest in *Phyla noduliflora* (19.162) of family Verbinaceae followed by *Acalypha indica* of family Euphorbiaceae and minimum in *Ageratum conyzoides* of family Compositae (1.245).

CONCLUSION

Heavy traffic passes through NH-31 (Katihar to Purnea). A large no. of truck, bus, car, auto, tractor and several other heavy vehicles continuously passes through this road. The pollutant released from these vehicles adversely affect herbs present at the road side. These pollutants causes decrease in chlorophyll content of herbaceous plants. In summer season, the atmosphere remain dry due to which pollutants remain within air for a long time. The vehicular pollutants mostly contain carbon monoxide, unburnt hydrocarbons, Sulphur and nitrogen oxide, tetraethyl lead and tetramethyl lead and several other lead compounds. These pollutants are absent in countryside so the chlorophyll content of the countryside herbs is higher than those of road side herbs.

Table 1- Comparison of Chlorophyll content in road side plants and plants one km away from road side

S. No.	Name of Plant	Family	Chl. Content of road side plant	Chl. Content of plant one km away from road side
1	*Catharanthus roseus*	Apocyanaceae	5.423	6.432
2	*Ageratum conyzoides*	Compositae	0.984	1.245
3	*Tridax procumbens*	Compositae	7.563	8.236
4	*Acalypha indica*	Euphorbiaceae	13.945	15.678
5	*Phyllanthus nururi*	Euphorbiaceae	10.645	12.421
6	*Parthenium hysterophorus*	Compositae	4.213	5.641
7	*Croton sparsiflorus*	Euphorbiaceae	8.456	9.898
8	*Euphorbia hirta*	Euphorbiaceae	4.321	5.814
9	*Emilia sonchifolia*	Compositae	8.974	10.573
10	*Phyla noduliflora*	Verbinaceae	17.961	19.162

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Comparative study of moisture content of ethnomedicinal plants used in Ranchi district of Jharkhand for curing chronic kidney injury

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Abstract- Jharkhand's Ranchi district has been a storehouse of vast natural resources, including vegetational abundance. The people that dwell in the area have their own traditions and taboos. Chronic kidney disease is the one of the major problem in India and all over the world. Some common plants like *Coriandrum sativum* L. and *Moringa oleifera* Lam. are used in this problem. The study was conducted to investigate moisture content in leaves of these two plants.

Key words: Moisture content, ethnomedicinal plants, chronic kidney disease, tribal communities.

INTRODUCTION

Throughout the history of human society, man and plant have had a very close relationship.[1] Ranchi is the capital of Jharkhand. Traditional treatments for various diseases are mostly plants used by the tribals of Ranchi district. Chronic Kidney Disease (CKD) is one of those diseases. Chronic kidney disease is a gradual loss over a period of months to years of kidney function. It is widely believed that once there has been sufficient initial damage, no matter what the cause, renal failure inevitably progresses.[2] The CKD may be associated with coexistent risk factors, such as hypertension, proteinuria, glycemia, anemia, and disturbances of lipid or mineral metabolism.[3]

The comparative study of moisture content of some plants, which are used in chronic kidney injury, was done. *Coriandrum sativum* L. is a medicinal and aromatic plant from the Apiaceae family that is found all over the world[4] and *Moringa oleifera* Lam. is a plant that belongs to the Moringaceae family and is considered under used. The plant is also known as drumstick, sahjan or sohanjana in India.[5] Both plants are used to cure chronic kidney injury. Indigestion, worm infections, rheumatism, loss of appetite, convulsions, sleeplessness, anxiety, and joint pain have all been treated using *Coriandrum sativum* L.[6] *Moringa oleifera* Lam. leaves are the most often used, and they have been shown to be effective in CKD as well as a number of other chronic conditions, such as hyper cholesterolemia, diabetes, hypertension, insulin resistance, non-alcoholic liver disease, and general inflammation.[7]

The moisture level of harvested plants is often high, ranging between 60 and 80 percent. If the moisture content is not considerably reduced, dangerous biological processes can mature. As a result, the beneficial active

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ingredient in herbs can be eliminated, and the medications' outward qualities can deteriorate.[8] The amount of moisture in a sample is expressed as a percentage of the sample's initial weight. The difference between initial weight of fresh leaves and final weight of same leaves after drying is the moisture content. The moisture content of each plant was estimated as a percentage, which differed from one to the next.

MATERIALS & METHODS

In study of moisture content, plants of two different family were harvested during early morning from the different areas of Ranchi district. The major blocks for collection of plants in Ranchi district has been shown in Fig. 1. Plants species were identified with the help of Botany of Bihar and Orissa.[9,10] The collected leaves were properly cleaned twice in running tap water to remove any dirt or surface contaminants that may have been present. The leaves were washed and then dried in a cold area before being weighted in the weighting machine. The leaves were then placed in a hot air oven at 55°C for half an hour. This activity was repeated 3-4 times. The dried leaves were then weighed. The difference between initial weight of the green leaves and final weight after drying was the moisture content. Percentage of moisture content has been calculated with this formula:

$$\text{Moisture content} = \frac{\text{Initial weight of leaves} - \text{final weight after drying}}{\text{Initial weight of leaves}} x\ 100$$

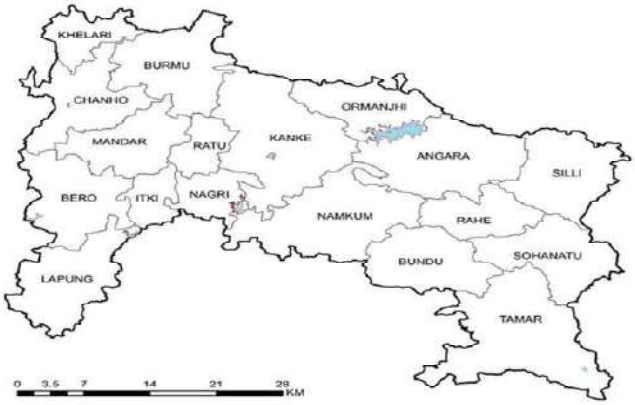


Fig. 1- Different blocks of Ranchi district for collection of plants.

RESULT & DISCUSSION

The comparative study of moisture content of two ethnomedicinal plants viz. *Coriandrum sativum* L. and *Moringa oleifera* Lam. has been done. *Coriandrum sativum* L. has a high moisture level, while *Moringa oleifera* Lam. has low moisture content as shown in table 1 & 2. The comparative study of weight of dried leaves (in gm) for *Coriandrum sativum* L. and *Moringa oleifera* Lam have been shown in Fig.2. It is found that *Moringa oleifera* Lam. has relatively high residue after drying. The *Moringa oleifera* Lam. plants have a lot of therapeutic properties as it has low average moisture content (69.91%) as shown in Fig.3. Their regular use can help prevent chronic renal disease. The *Moringa oleifera* Lam. plants can be used in our daily life to improving the filtration rate of kidney.

Table 1: Moisture content of *Coriandrum sativum* L.

Material	Number of observation	Weight of fresh leaves (in gm)	Weight of dried leaves (in gm)	Difference (in gm)	% of Moisture content
Fresh leaves of *Coriandrum sativum* L.	1.	10	2.02	7.98	79.8%
	2.	10	1.96	8.04	80.4%
	3.	10	1.96	8.04	80.4%
	4.	10	1.94	8.06	80.6%
	5.	10	1.94	8.06	80.6%
	6.	10	1.98	8.02	80.2%
	7.	10	1.88	8.12	81.2%
	8.	10	1.82	8.18	81.8%
	9.	10	1.86	8.14	81.4%
	10.	10	1.82	8.18	81.8%
Total					**808.2%**

Average Moisture content = 80.82%